

Circulatory System

Circulatory System:-

This is the transport system which circulates blood and lymph to and from the cells throughout the body to provide nourishment and maintain homeostasis.

Homeostasis = (Stead internal condition)

There are two types of circulations as

- **Systemic Circulation**
- **Pulmonary circulation**

Systemic Circulation:-

This circulation supplies blood throughout the body except lungs.

Pulmonary Circulation:-

This is the circulation of blood to and from the lung.

Heart

[Pumping organ]

Vessels

[Transport medium]

Systemic circulation in heart

Left ventricle to whole body via aorta and back to right atria via vena cava.

Pulmonary Circulation in heart

Right ventricle to lungs via pulmonary artery and back to left atria via pulmonary vein.

There are two types of valves in heart as

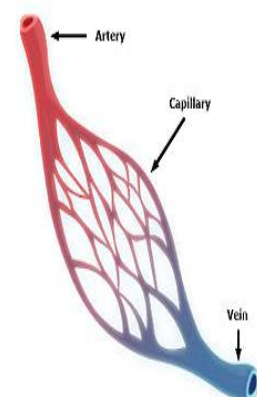
- **Tricuspid, Mitral valves/Bicuspid)**
- **Semi-lunar valves (Aortic/Pulmonary)**

Mediastinum:-

It is loose connective tissue that contains a group of structure within thorax region.

Properties:-

- Hollow (About the size of a closed fist)
- Muscular (225g)
- Outer membrane (Pericardium)



- Blood supply to heart via **Coronary Arteries**

Functional parts of circulation

- **Heart:-**

This is a conical, hollow, muscular organ situated in the middle of mediastinum. It is enclosed within pericardium.

It pumps blood to various parts of body to meet their

Respective nutritive requirements.

- **Arteries:-**

These are a series of blood vessels which carry oxygenated blood away from the heart. The walls of arteries are constructed of three coats of tunica. They have a hollow core called **LUMEN** through which blood flows.

- **Arterioles:-**

These are the smaller branches of arteries which deliver blood to capillaries.

- **Capillaries:-**

These are diffuse network of blood vessels which connect the arterioles and venules. Capillaries are found in almost every cell body. The distribution of capillaries in the body varies with the activity of the tissues.

- **Venules:-**

These are blood vessels that connect capillaries with veins.

- **Veins:-**

These are a series of blood vessels which carry blood towards the heart (except pulmonary and umbilical veins). Veins are composed of essentially the same three coats as arteries, but there are variations in their relative thickness.

There are three layers of artery as

- The outer layer is called **Tunica Externa**. It is made of Collagen and Elastic fiber.
- The middle layer is called **Tunica Media**. It is made of smooth cells.
- The inner layer is called **Tunica Interna**. It is made of Endothelial cells.

Syncytium:-

A single cell or cytoplasmic mass containing several nuclei, formed by fusion of cells or by division of nuclei.

Properties of Cardiac Muscle

- **Nature:-**

Cardiac muscle is involuntary in function and striated in appearance.

- **Cardiac muscle as a syncytium:-**

When the number of cell function as a single unit, the whole functional unit is called Syncytium.

The heart is composed of two different syncytia, the atrial syncytium and ventricular syncytium.

- **Automaticity/rhythmicity:-**

It is the ability of the cell to undergo depolarization.

Rhythmicity denotes spontaneous depolarization occurring at regular interval, as Na^+ ions are leaky to S.A node fibers that cause depolarization (action potential) at regular interval.

- **Conductivity:-**

This is the transfer of cardiac impulses from the Sino-atrial node (S.A node) to all the cardiac muscle.

- **Pathway:-**

This phenomenon occurs through a specialization conducting system, consisting of S.A node, internodal pathway, A.V node, A.V.

- **Contractility:-**

This is the contraction of cardiac muscle due to interdigitation of actin & myocin filament in the presence of ATP (Adenosine triphosphate) and calcium ion.

- **Refractory period:-**

This is the time period during which an already excited muscle cannot be re-excited by a normal stimulus.

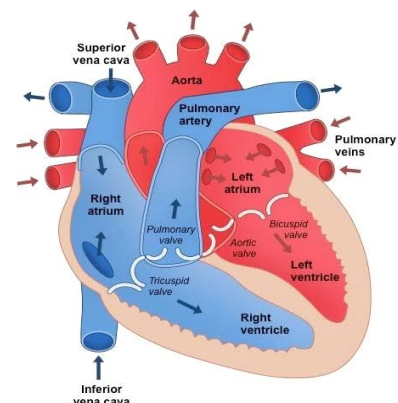
Refractory period is of two types

- Absolute refractory period
- Relative refractory period
- **All or nothing law:-**

This cardiac muscle fiber do not contract if the stimulus is subthreshold. They will contract with constant strength if the stimulus is at or above the threshold.

Cardiac Cycle

Cardiac Cycle:-



Cardiac cycle is the performance of human heart from the beginning of one heart beat to the beginning of next heartbeat.

Or

Events taking place in human heart in each beat.

Contraction (Systole)

Heart

Relaxation (Diastole)

Events of Cardiac Cycle

Events of cardiac cycle is classified into: **Atrial event and ventricular event**

- **Atrial events:-**

- Atrial systole (0.1 sec to contract)
- Atrial diastole (0.7 sec to contract)

Atrial systole is also known as rapid filling phase. During this period right and left atria contract and blood is forced to move from atria to ventricles. Only 10% blood is pumped from atria to ventricles during atrial systole. The duration of this phase is 0.11 seconds.

Atrial diastole is the phase in which atria relax and during this period atrial filling take place. Right atria receive deoxygenated blood from all over the body through superior and inferior vena cava. Left atrium receive oxygenated blood from lungs through pulmonary veins. The duration of this phase is 0.69 seconds.

- **Ventricular event:-**

- Ventricular systole (0.3 sec to contract)
- Ventricular diastole (0.5 sec to contract)

- **Ventricular systole:-**

Ventricular systole is the contraction of ventricles during which right ventricle pumps deoxygenated blood to lungs through pulmonary artery via pulmonary valve (pulmonary circulation) and left ventricle pumps oxygenated blood to the body through aortic artery via aortic valve (systemic circulation)

Ventricular systole completes in two steps:

- **Isometric contraction:-**

Isometric contraction of ventricular muscle is also called isovolumetric contraction. During isometric contraction ventricles contract in such a way that there is no change in blood volume in ventricles because all valves (mitral and semilunar) are closed and pressure in ventricles is increased.

- **Ejection period:-**

Increase in pressure in ventricles causes opening of semi-lunar valves. Blood is ejected out of both ventricles through these valves hence this period is called ejection period.

- **Ventricular diastole:-**

It is the relaxation of ventricles in which right ventricle receive deoxygenated blood from right atria via tricuspid valve and left atria receive oxygenated blood from left atria via bicuspid valves. This phase is complete in 0.053 seconds.

Ventricular diastole completes in five steps:

- **Protodiastole:-**

Protodiastole is the first stage of ventricular diastole. No other change occur in the heart during this period.

- **Isometric relaxation:-**

Isometric relaxation of ventricle muscle is also called isovolumetric relaxation. During isometric relaxation ventricles relax in such a way that there is no change in blood volume in ventricles because all valve (mitral and semi-lunar) are closed and pressure in ventricles is decreased. Decreased in pressure in ventricles causes opening of semi-lunar valves.

- **Rapid filling phage:-**

Opening of Atrio-ventricular valves causes a sudden rush of blood from atria into ventricles. So, this period is called the first rapid filling.

- **Slow filling phage:-**

After the sudden rush of blood the ventricles filling becomes slow. Now it is called slow filling.

- **Last rapid filling phage:-**

Last rapid filling occurs because of atrial systole. The atria contract and push a small amount of blood into ventricles i.e 10%. Flow of additional amount of blood in ventricle due to atrial systole is called atrial kick.

Cardiac cycle involves four stages

- Isovolumetric Relaxation
- Inflow
- Isovolumetric contraction
- Ejection

Heart Sounds

Introduction:

Heart sounds are heard with the help of a stethoscope (discovered by Lenic). Two sounds are normally heard during the cardiac cycle. In some cases, one may hear a 3rd or 4th heart sound. These sounds are made by the closure of the heart valves and the acceleration and deceleration or vibration of valves due to blood flow in the cardiac chambers.

Heart sounds:

Heart sounds are the sound produce by mechanical activities of heart during each cardiac cycle.

Mechanical Activities

- Closure of valves of the heart
- Contraction of cardiac muscle
- Flow of blood through chambers

There are four types of heart sounds:

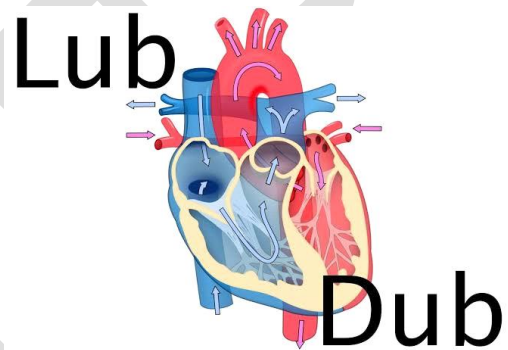
- **1st heart sound:**

The first heart sound is produce during isometric contraction period due to closure of Atrio-ventricular (Tricuspid/Bicuspid) valves and its sound is like “LUBB”.

- **2nd heart sound:**

Second heart sound is produce at the start of isometric relaxation period due to closure semi-lunar (Aortic/pulmonary) valves and its sound like “DUBB”.

- **3rd heart sound:**



It is low pitched heart sound produce during rapid filling period of cardiac cycle.

- **4th heart sound:**

It is inaudible and produce during atrial systole.

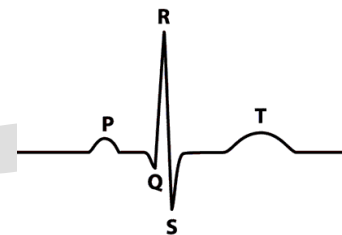
Murmurs:

These are abnormal heart sounds produced when there is an excessive degree of turbulence flow in the heart chamber.

Electrocardiogram (ECG)

Electrocardiography is the technique of recording the Electrical activity of the heart using electrodes.

Electrocardiography is a machine.



There are three types of waves:

- **P – wave**
- **QRS complex (Q wave, R wave, S wave)**
- **T - wave**

Intervals:

- PR – interval (Atrial Depolarization)
- QRS – interval (Ventricular Depolarization)
- QT – interval (ventricular Depolarization & Ventricular Repolarization)
- ST – interval (Ventricular Repolarization)

Depolarization:

It is a change which undergoes a shift in electrical charge being less negative inside is called depolarization.

Repolarization:

It is a change which undergoes a shift in electrical charge being more negative inside is called repolarization.

P–wave => Depolarization => contraction => systole => Atrial-systole

QRS–wave => Depolarization => contraction => systole => Ventricular systole

T-wave => Repolarization => relaxation => diastole => Ventricular diastole

Blood Pressure (BP)

(Arterial blood pressure)

Artery

Vessel

The pressure exerted by blood on the walls of arteries is called **blood pressure**.

There are four types of blood pressure:

- Systolic blood pressure
- Diastolic blood pressure
- Pulse pressure
- Mean arterial blood pressure

Systolic Blood Pressure:

The maximum pressure exerted by blood in the artery during systole of heart is called systolic blood pressure. Its value is (120mm/Hg).

Diastolic Blood Pressure:

The minimum pressure exerted by blood in the artery during diastole of heart is called diastolic blood pressure. Its value is (80mm/Hg).

Pulse Pressure:

The difference between systolic and diastolic blood pressure is called pulse pressure. Its value is (40mm/Hg).

Mean Arterial Blood Pressure:

Mean arterial blood pressure is the diastolic pressure plus one third of the pulse pressure.

Diastolic pressure + 1/3 of pulse pressure

$$80 + 40/3 = 93.3 \text{ mm/Hg}$$

Stroke Volume:

The amount of blood put out by the left ventricle of the heart in one contraction is called stroke volume.

Cardiac output:

The amount of blood the heart pumps through the circulatory system in one minute is called cardiac output. We can also say it systolic blood pressure.

A normal adult has a cardiac output of 4.7 liters (5 quarts) of blood per minute.

Determinants of arterial blood pressure – Factor Maintaining Arterial Blood Pressure

Some factors are necessary to maintain normal blood pressure. These factors are called **local factors**, **mechanical factor** or determinants of blood pressure.

Types of local factors

Local factors are divided into two types:

- **Central factor**

Central factors are pertaining to the heart:

- Cardiac output
- Heart rate

Cardiac factors:

Cardiac output:

Systolic pressure is directly proportional to cardiac output. The cardiac output increases systolic pressure increases and when cardiac output is less, the systolic pressure is reduced.

Heart rate:

Moderates change in heart rate do not affect arterial blood pressure much. However, marked alteration in the heart rate affects the blood pressure by altering cardiac output.

- **Peripheral Factors**

Peripheral factors are pertaining to blood and blood vessels.

- **Peripheral Resistance:**

It is important factor, which maintains diastolic pressure. Diastolic pressure is directly proportional to peripheral resistance. Peripheral resistance is the resistance offered to the blood flow at the periphery.

- **Blood volume:**

Blood pressure is directly proportional to blood volume. Blood volume maintains the blood pressure through the venous return and cardiac output. If the blood volume increases, there is an increase in blood volume and cardiac output.

- **Venous return:**

Blood pressure is the directly proportional to the venous return. When venous return increases, there is an increase in ventricular filling and cardiac output.

- **Elasticity of blood vessels:**

Blood pressure is inversely proportional to the elasticity of blood vessels.

- **Velocity of blood flow:**

Pressure in blood vessel is directly proportional to the velocity of blood flow. If the velocity of blood flow increases, the resistance is increased. So pressure is increased.

Physiological variations of blood pressure

Diurnal variation:

Blood pressure is lowest early in the morning and highest in the evening.

Age:

Blood pressure rises with age. The systolic blood pressure at different ages:

Infancy = 80-90 mm/Hg

Childhood = 90-110 mm/Hg

Adult 110 =120 mm/Hg

Elderly = 140-150 mm/Hg

Sex:

Before the occurrence of menopause in the females both the systolic and diastolic blood pressures are slightly lower than males. After menopause blood pressure may be slightly higher than males of the same age.

Body build:

Obese persons tend to have higher blood pressure.

Sleep:

In resting condition as in sleep, blood pressure is decreased.

Exercise:

Light exercise such as walking increases the systolic blood pressure, but decreases the diastolic blood pressure.

In severe exercise the systolic blood pressure may rise up to 180 mm/Hg.

Posture:

Standing causes an increase in both systolic and diastolic blood pressure.

Excitement or emotion:

This may cause an increase in systolic blood pressure.

After digestion of food:

There is a slight rise in systolic blood pressure, but the diastolic blood pressure falls due to vasodilation in certain vessels of body.